

2920/202A
COMPUTER APPLICATIONS II (THEORY)
PAPER 1
November 2021
Time: 2 hours



THE KENYA NATIONAL EXAMINATIONS COUNCIL
DIPLOMA IN INFORMATION COMMUNICATION TECHNOLOGY
MODULE II
COMPUTER APPLICATIONS II (THEORY)

Paper 1

2 hours

INSTRUCTIONS TO CANDIDATES

This paper consists of SIX questions.

Answer any FOUR of the SIX questions in the answer booklet provided.

Candidates should answer the questions in English.

This paper consists of 4 printed pages.

Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.

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Turn over

3 (a) With the aid of an example, explain the meaning of the term *machine learning* as used in artificial intelligence. (3 marks)

(b) Explain the use of artificial intelligence as applied in each of the following areas: (4 marks)

(i) natural language processing;

(ii) gaming.

(c) Distinguish between *spatial data* and *attribute data* as used in a Geographical Information System. (4 marks)

(d) The balances in the commission income account of Pesa Agents is as indicated.

Date	01.01.2020	31.12.2020
Balance due (Ksh)	45,260	85,200

The business received Ksh 507,600 as commission during the year. Show the necessary entries in the commission income account of Pesā Agents for the year 2020. (4 marks)

4. (a) Outline **three** main components of Geographic Information Systems. (3 marks)
- (b) State the sequence of Computer-Aided Design commands to draw a line with starting point coordinates of 0,0 and terminating 8 units on the X axis and 5 units on the Y axis using the absolute coordinate system. (3 marks)

- (c) Describe each of the following systems as used in artificial intelligence:
- (i) Neural networks; (2 marks)
- (ii) Genetic algorithm. (2 marks)

- (d) The final balances relating to Jalani Traders for the month of June 2020 are as follows:

Item	Ksh
Credit Purchases	11,405,230
Cash Purchases	9,852,410
Opening stock	487,920
Credit Sales	28,745,620
Cash sales	4,895,210
Returns outwards	415,620
Returns inwards	197,300
Gross profit	11,316,290
Drawings Stock	81,900
Drawings cash	870,200

Determine the closing stock for the business

- 5 (a) Explain the term *deep learning* as used in artificial intelligence. (2 marks)

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Turn over

$20,21,250$
 $11,316,290 = X - 33,225,210$
 $X = 44,541,500 -$

$$11,316,290 = 21,548,260 - 33,225,210$$

- (b) Outline **two** types of each of the following attribute data in a Geographic Information Systems: (6 marks)
- (i) Categorical data;
 - (ii) Numeric data.
- (c) The following is a list of application software:
Adobe InDesign, CorelDraw, FreshBooks, Intacct, Microsoft Publisher, Notepad++, QuickBooks, Quicken, Sage50, Virtual C, Wavepad, Xero, Zoho Books
 Identify **six** accounting software from the list. (3 marks)
- (d) When a transaction takes place, it is first recorded in a book of original entry. Outline **four** such books used in accounting. (4 marks)
- 6 (a) Outline **four** types of precision errors associated with information from a Geographic Information System. (4 marks)
- (b) A Computer-Aided Design user has a drawing which is to be modified. Outline **six** commands in the program that he is likely to use. (3 marks)
- (c) Outline **three** factors to consider when selecting a *search algorithm* to be used in artificial intelligence. (3 marks)
- (d) With the aid of an example, distinguish between *accounts payable* and *accounts receivable* as used in accounting. (5 marks)

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